MATH 351 Midterm 1

Fall 2025

Name ___________________________________

Rules of the test

- Time limit: test is due at the end of our usual class time. I have written the test expecting you to finish within 85 minutes.
- There are four problems. I will weight them equally.
- All four problems have multiple parts. You may be able to complete later parts successfully even if you have not completed earlier parts successfully.
- Read each problem carefully. Many details in what I have written are important to the interpretation of the problems. Give clear explanations using the context of the problem.
- If you are uncertain about what to do with any of the questions on the test, please ask me so that I can point you in the right direction.
- The exam is to be taken individually—no collaboration of any kind is allowed.
- Here is the allowed aid for this test. You may use a single-sided handwritten sheet of notes during the test. You will submit your sheet of notes with your completed exam. You may also use a calculator on all questions. You may not use your phone as a calculator. If you need to borrow a calculator, I can provide one.
- If you need to leave the room during the exam, you must leave your phone with me.
- If I have reason to suspect inappropriate collaboration or use of aid that I have not allowed for this test, I will give everyone involved a score of 0.

Pledge

When you are done, come back here to sign the honor pledge.

I pledge that I have neither given nor received any unauthorized aid on this exam.

*Signed:*___________________________

1. This first question is about terminology and general principles in probability theory. In all the questions, (Ω, P) is a probability model with sample space Ω and probability function P (assigning a probability to each event).

a. What exactly is a random variable for (Ω, P)? I am looking for the "pure math" definition of "random variable"—not specific examples or the informal "real world" meaning—but write that if it is all you know.

A random variable is a function that takes outcomes in the sample space Ω as inputs and gives numbers as outputs.

b. Suppose that A, B, C are events in Ω. Use the algebra of events to express the event "at least one of A, B, C happens" in terms of A, B, C.

$A \cup B \cup C$

c. Use the algebra of events to express "A happens and B happens but C does not happen" in terms of A, B, C.

$A \cap B \cap C^c$

d. When do we add probabilities of events? When do we multiply probabilities of events? Give the general principle and an example of each situation.

Principle for when we add:

We add probabilities when calculating $P(A \cup B)$, where A and B are disjoint events. In this case, we have $P(A \cup B) = P(A) + P(B)$.

Example:

Our chance experiment is rolling a die. A = roll a 2 on the die. B = roll a 3 on the die. These events are disjoint, and $P(A \cup B) = P(A) + P(B)$.

Principle for when we multiply:

We multiply probabilities when calculating $P(A \cap B)$, where A and B are independent events. In this case, we have $P(A \cap B) = P(A)\, P(B)$.

Example:

Our chance experiment is rolling a die twice. A = get a 2 on the first roll. B = get a 3 on the second roll. These events are independent, and $P(A \cap B) = P(A)\, P(B)$.

2. In a chance experiment, we will flip a coin 5 times.

a. Describe the sample space Ω with equally likely outcomes that you would use for modeling this experiment.

We could model the experiment using outcomes that are sequences of five H's and T's. The set of all such sequences is the sample space.

b. How many outcomes are there in your sample space?

There are $2^5 = 32$ outcomes in this sample space. We can count using the multiplication rule, making a sequence of H's and T's by choosing one at a time. There are 2 choices at each stage and thus (2)(2)(2)(2)(2) = 32 elements in the sample space.

Let *Z* be the random variable giving the difference between the number of heads flipped and the number of tails flipped. The difference can be negative! For example, for an outcome where you get 2 heads and 3 tails, the value of *Z* is 2 – 3 = –1.

c. What are the possible values of *Z*?

Here are the possibilities for the numbers of heads and tails flipped and the corresponding values of Z:

0 H and 5 T: Z = –5

1 H and 4 T: Z = –3

2 H and 3 T: Z = –1

3 H and 2 T: Z = 1

4 H and 1 T: Z = 3

5 H and 0 T: Z = 5

d. Find the distribution of the random variable *Z* in your model, i.e. give the probability of each possible value of *Z*.

We can use the binomial(5, 1/2) distribution to calculate the probabilities P(Z = k) for the values of k listed above.

P(Z = –5): This event corresponds to 0 H and 5 T. The probability is $C_{5,0}\,(1/2)^0(1/2)^5 = 1/32$

P(Z = –3): This event corresponds to 1 H and 4 T. The probability is $C_{5,1}\,(1/2)^1(1/2)^4 = 5/32$

P(Z = –1): This event correspond to 2 H and 3 T. The probability is $C_{5,2}\,(1/2)^2(1/2)^3 = 10/32$

P(Z = 1): This event correspond to 3 H and 2 T. The probability is $C_{5,3}\,(1/2)^3(1/2)^2 = 10/32$

P(Z = 3): This event corresponds to 4 H and 1 T. The probability is $C_{5,4}\,(1/2)^4(1/2)^1 = 5/32$

P(Z = 5): This event corresponds to 5 H and 0 T. The probability is $C_{5,5}\,(1/2)^5(1/2)^0 = 1/32$

e. I hope that the expected value of *Z* in your model is 0! What is its variance?

Since $E[Z] = 0$, $Var(Z) = E[Z^2]$.

$E[Z^2] = (25)(1/32) + (9)(5/32) + (1)(10/32) + (1)(10/32) + (9)(5/32) + (25)(1/32) = 160/32 = 5$

3. Our class has 15 people in it.

a. I will pick three of the 15 students in our class to give prizes. (First prize is a delicious apple, second prize is my last summer tomato, and third prize is a ripe pear.) How many different ways could I distribute the prizes? Explain your counting method briefly, with one or two sentences.

We are picking a permutation of three students from the 15 students in the class. The number of such permutations is $P_{15, 3}$ = (15)(14)(13) = 2,730.

We could also go back to the multiplication rule: We first pick a student to get the apple (15 choices). Then we pick a student to get the tomato; not matter who gets the apple, there are 14 choices left. Finally, we pick who gets the pear; no matter who gets the apple and who gets the tomato, there are 13 choices left for the pear. Therefore, by the multiplication rule, there are (15)(14)(13) ways of giving out the prizes.

b. Imagine that I pick three winners from the 15 people in the class by drawing names with replacement from a hat: after I pick a name for one prize, I put the name back in the hat. This means that one person could get more than one prize. How many ways could I give out the three prizes to the 15 people in class? Explain your counting method briefly, with one or two sentences.

We use the multiplication rule. There are 15 possibilities for who gets the apple, 15 possibilities for who gets the tomato, and 15 possibilities for who gets the pear. Therefore, by the multiplication rule, there are (15)(15)(15) = 3,375 total possibilities.

c. If I draw names with replacement to assign the prizes, what is the probability that one student gets at least two prizes? Show enough of your calculations below so that I can understand what you did. Please also use a calculator to find a decimal approximation to this probability. When I calculated this probability, I got about 0.2.

Solution 1: From part b, there are (15)(15)(15) ways that I could pick three names from the hat. From part a, (15)(14)(13) of those ways have three distinct names. Therefore, (15)(15)(15) – (15)(14)(13) ways have at least one repeated name. The probability of at least one repeated name is therefore $(15^3 – (15)(14)(13)) / 15^3$ or about 0.191. This is how we would have solved the problem at the start of the course. It is the easier solution.

Solution 2: You could also solve the problem in a more complicated way using our new tools. Think about a single student (maybe you!). The number of prizes that student gets in a random draw of three names has the binomial(3, 1/15) distribution. The probability that this student gets at least two prizes is therefore $C_{3,2}\,(1/15)^2(14/15) + C_{3,3}\,(1/15)^3(14/15)^0$. There are 15 students in the class. The events “student i gets at least two prizes” (for i = 1, 2, ..., 15) are disjoint. Therefore, to calculate the probability that at least one student gets at least two prizes, we can add together 15 copies of the probability that a single student gets at least two prizes. That works out to be $15(C_{3,2}\,(1/15)^2(14/15) + C_{3,3}\,(1/15)^3) \approx 0.191$.

4. The second Earl of Yarborough is reported to have bet at odds of 1000 to 1 that a hand of 13 cards (out of a standard deck of 52) would contain at least one card that is ten or higher. By "ten or higher" we mean that a card is either a ten, jack, queen, king or ace.

a. Show how to calculate the probability that the earl loses this bet. Please report your probability in exact form, explaining briefly how you calculated it. You do not need to calculate a decimal value.

This is an "urn problem." In the deck of 52 cards, there are 20 cards whose value is ten or higher and 32 cards whose value is less than ten. The probability that the earl's hand of 13 cards has k cards (for k = 0, 1, 2, ..., 13) with value ten or higher is

$$\frac{C_{20,k}C_{32,13-k}}{C_{52,13}}.$$

Taking k = 0, we get

$$P(\text{no cards ten or higher})=\frac{C_{20,0}C_{32,13}}{C_{52,13}}.$$

b. Using a calculator, I got about 0.000547 for the probability that the earl loses the bet. Saying that he bet 1000-to-1 odds means that if he wins the bet, he earns \$1 and that if he loses the bet, he pays \$1000. What is his expected value for the bet?

The expected value is (\$1)P(earl wins) + (–\$1000)P(earl loses). That works out to be

$$(\$1)(1-0.000547)+(-\$1000)(0.000547)=\$0.452453.$$

c. Imagine that in a crazy weekend at whatever parties such English nobles attend, the earl makes this bet 100 times. Let *X* be the random variable counting the number of times that he loses the bet out of these 100 trials. What distribution would you use for *X*? What is *E[X]*, the expected number of times the earl will lose during the weekend's play?

$X \sim$ binomial(100, 0.000547)

$E[X]$ = (100)(0.000547) = 0.0547, according to the formula for expected value of a binomial random variable.

d. The earl brought \$1000 to the party and will thus be ruined if he loses two times or more. Calculate (or estimate) the probability of this disaster. Show enough work that I can understand your methods.

$P(X = 0) = C_{100,0}\,(1 - 0.000547)^{100} \approx 0.9467549669$

or use Poisson approximation $e^{-0.0547} \approx 0.9467691361$

$P(X = 1) = C_{100,1}\,(0.000547)^{1}(1 - 0.000547)^{99} \approx 0.05181583996$

or use Poisson approximation $0.0547\ e^{-0.0547} \approx 0.05178827174$

P(at least two losses in 100 bets) = 1 – P(X = 0) – P(X = 1) ≈ 0.001429193104 (binomial) or 0.001442592162 (Poisson approximation)